

Question	Marking Guidance	Mark	Comments
09.1	Correct answer in range 238 to 240 = 2 marks;; OR In working shows: 16 (ignoring preceding zeros or subsequent numbers and position of decimal point e.g. 0.016) = 1 mark OR 10909 or 10910 (ignore subsequent numbers and position of decimal point) = 1 mark OR 1.00656 (ignore subsequent numbers and allow correct rounding) = 1 mark;	2 (2 x AO2)	

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09.2	1. (Short) single strand of DNA; 2. Bases complementary (with DNA allele/gene);	2 (2 x AO1)	2. Accept complementary base sequence

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09.3	1. and 2. 1 mark per correct column;; <table><tr><th>Lane</th><th>Parent or child</th><th>Genotype</th></tr><tr><td>1</td><td>Parent</td><td>Ff / fF</td></tr><tr><td>2</td><td>Child</td><td>ff</td></tr><tr><td>3</td><td>Parent</td><td>Ff / fF</td></tr><tr><td>4</td><td>Child</td><td>FF</td></tr></table> 3. (Lane 2 only contains) recessive/f allele which travels further due to fewer <u>bases</u> OR (Lane 2 only contains) recessive/f allele that travels further due to deletion of (3) <u>bases</u>;	Lane	Parent or child	Genotype	1	Parent	Ff / fF	2	Child	ff	3	Parent	Ff / fF	4	Child	FF	3 (3 x AO3)	1 and 2. Accept if columns are reversed 2. Accept heterozygous, homozygous recessive, heterozygous and homozygous dominant 2. Accept use of different letters other than F and f 3. Accept 'triplet' for (3) bases 3. Accept F508del or mutation for recessive allele 3. In this instance accept gene for allele
Lane	Parent or child	Genotype																
1	Parent	Ff / fF																
2	Child	ff																
3	Parent	Ff / fF																
4	Child	FF																
